

Lead carboxylates passivation for metre-scale perovskite solar modules

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Environmentally sensitive ammonium halides have emerged as indispensable surface passivators for perovskite solar cells, demonstrating broadly effective performance at the laboratory scale^{1–4}. However, the sensitivity of ammonium halides to humidity typically necessitates operation under an inert atmosphere⁵. Coupled with the heterogeneous distribution of passivators during large-area slot-die coating, applying such surface treatment techniques to the fabrication of metre-scale perovskite solar modules (PSMs) under ambient conditions remains challenging. Here we instead use chemically stable lead carboxylates as passivators on perovskite films engineered to have a naturally formed formamidinium iodide (FAI)-enriched surface by means of a high-saturated vapour pressure (SVP) solvent system (2-methoxyethanol (2-Me), 1,3-dioxolane (DOL) and dimethyl sulfoxide (DMSO)). By treating the tailored surface with lead dioleate (Pb(OA)₂) to enhance carrier transport and surface passivation, we fabricated PSMs that achieved certified efficiencies of 24.0% (aperture area of 810 cm²) and 22.0% (total area 0.72 m²) and passed all International Electrotechnical Commission (IEC) 61215 reliability tests. These results represent the highest reported performance for scalable, industrially viable perovskite photovoltaics (PVs).

PVs, renowned for their high efficiency^{1–4} achievable through cost-effective solution-based fabrication techniques^{6–8}, are now advancing from laboratory-scale research to industrial-scale manufacturing^{9–12}. In this transition, controlling perovskite film formation over large areas is critical and typically achieved by slot-die coating perovskite solution followed by vacuum chamber drying (VCD)^{9,11}. The low-pressure-driven upward solvent extraction of VCD together with the varying solubility of perovskite precursors determine the vertical compositional grading in films, that is, where some ingredients accumulate at the buried interface or the top surface^{13,14}. In other words, the solvent system used for perovskite deposition plays a decisive role not only in controlling the crystallization kinetics over large areas but also in vertical compositional distribution and, hence, surface chemistry and, ultimately, device performance.

The widely used solvents such as N,N-dimethylformamide (DMF) and DMSO exhibit low SVP¹⁰, which unfortunately hinders controlled graded compositional distribution and regulated solvent-evaporation kinetics under VCD conditions. On the other hand, high-SVP solvents, such as 2-Me, promote rapid directional solvent escape and surface enrichment of more-soluble components. However, the 2-Me alone fails to solubilize Cs-containing perovskite inks^{15,16} and therefore needs to be blended with low-SVP cosolvents, such as DMSO; even

then, unfortunately, the fundamental mechanisms behind its failure to maintain perovskite ink stability remain unexplained and unresolved. Although several solvent engineering strategies have been explored to regulate perovskite crystallization using the VCD technique, the interplay between evaporation kinetics and the resulting component distribution at the microscopic scale remains ambiguous.

The surface composition formed in perovskite film growth as a result of vertical compositional grading determines the choice of surface passivator, which has been instrumental for improving perovskite PV performance. Ammonium halides passivators (AHPs) are routinely used for this purpose, as they form low-dimensional perovskites or defect-passivating layers that initially improve the efficiency^{17–21}; however, on exposure to environmental stressors such as moisture, heat or light, including during ambient-processed fabrication, the AHPs undergo deprotonation, leading to a progressive loss of performance^{22–29}. Alternative passivators based on organic polymers or small molecules with carboxyl, sulfhydryl, sulfonyl, amine, aromatic rings and halogen groups bind weakly to perovskite surfaces and hence suffer from photothermal instability^{30,31}. Non-reactive passivators based on inorganic salts (for example, PbSO₄ (ref. 32), Pb(SCN)₂ (ref. 33), MoO_x (ref. 34), LiF (ref. 35)) by means of ionic or covalent bonds often yield an uncontrollable thickness, impeding charge transport across the

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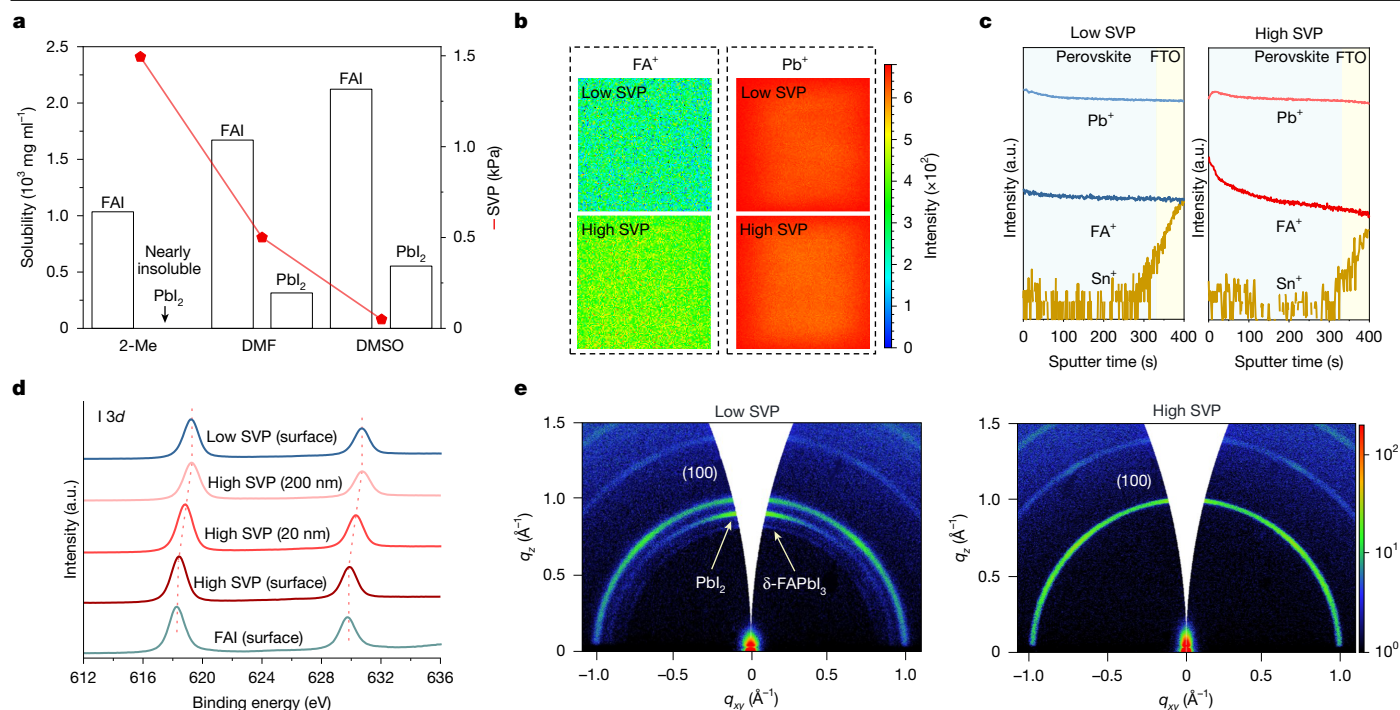


Fig. 1 | Surface states of perovskite films. **a**, Solubility of FAI and PbI_2 in 2-Me, DMF and DMSO and the SVP of these solvents at room temperature. **b**, TOF-SIMS surface ion mapping of the FA^+ and Pb^+ on the low-SVP and high-SVP prepared perovskite films. **c**, TOF-SIMS depth profile of the perovskite films. **d**, I 3d XPS

spectra of pristine FAI and perovskite films prepared from low-SVP and high-SVP solvent systems. **e**, GIWAXS patterns of the perovskite films prepared by low and high SVP. a.u., arbitrary units.

interface. Lead carboxylates passivators (LCPs) have emerged as a promising alternative owing to their physicochemical stability³⁶. We posit that designing a perovskite surface amenable to a controlled reaction with a LCP can yield a chemically bonded, well-defined passivation layer that maintains efficient charge extraction while providing robust environmental protection.

Here we reveal the relationship between the solvent-evaporation kinetics under VCD and the microscopic distribution of perovskite components. Specifically, we design a FAI-enriched perovskite surface through a high-SVP solvent system, composed of 2-Me, DOL and DMSO, which then, on reaction with a LCP, $\text{Pb}(\text{OA})_2$, forms an atmospherically robust barrier. Corresponding 810-cm^2 (aperture area) and 0.72-m^2 (total area) PSMs achieve certified power conversion efficiencies (PCEs) of 24.0% and 22.0%, respectively. These modules also demonstrated full compliance with the IEC 61215 standards.

Surface composition control via solvent chemistry

The VCD imposes specific requirements on the SVP of the solvents used. Common perovskite ink solvents, such as 2-Me, DMF and DMSO, exhibit sequentially increasing solubility for FAI and lead iodide (PbI_2), whereas their SVP decreases in the same order (Fig. 1a and Supplementary Table 1). During VCD, the chamber pressure is typically maintained below 10 Pa, leading to the rapid evacuation of high-SVP 2-Me, the gradual removal of low-SVP DMF and DMSO, ultimately resulting in the formation of a wet perovskite film⁸. In the low-SVP ink system, in which DMSO acts as the coordinating solvent and DMF as the primary solvent, the ink exhibits slow evaporation and high solubility for FAI and PbI_2 . We speculate that, although the high solubility of FAI is maintained in this system, its diffusion rate is kinetically sluggish and necessitates high-temperature treatment, thus often resulting in an incomplete reaction with PbI_2 at the surface during film formation.

High-SVP solvents such as 2-Me allow for surface customization using VCD. However, the 2-Me causes cloudiness and precipitation

of Cs-containing perovskite solution, thereby preventing the formation of stable ink^{16,37}. We introduced DOL to stabilize perovskite ink by means of cosolvency with 2-Me:DOL, thus establishing a high-SVP ternary solvent system of 2-Me:DOL:DMSO. With the incorporation of low-boiling-point, low-polarity DOL, Cs^+ preferentially coordinates with the oxygen atoms in the 2-Me:DOL complex instead of incorporating into the PbI_2 lattice, thereby effectively stabilizing Cs-containing perovskite ink (Supplementary Note 1 and Supplementary Figs. 1–11).

When using the high-SVP 2-Me:DOL as the primary perovskite solvent, we found that a FAI-enriched surface was formed on perovskite film. Time-of-flight secondary ion mass spectrometry (TOF-SIMS) surface ion mapping revealed FA^+ accumulation for the high-SVP perovskite film, in contrast to the Pb^+ accumulation for the low-SVP perovskite film (Fig. 1b). The depth profile ion distribution also demonstrated FA^+ accumulation at the top surface of the high-SVP perovskite film (Fig. 1c and Supplementary Fig. 12). X-ray photoelectron spectroscopy (XPS) showed that the high-SVP perovskite surface exhibits binding energies (I 3d, N 1s) nearly identical to bare FAI, whereas its bulk characteristics resemble those of the low-SVP perovskite (Fig. 1d and Supplementary Fig. 13). This indicates that the FA^+ and I^- ions reside in an essentially uncoordinated state at the surface.

The formation of the FAI-enriched surface is attributed to the vacuum-induced directional migration of solvent from the film interior towards the upper surface, which drives convective transport and facilitates the accumulation of solute (FAI) on the wet perovskite film (Supplementary Fig. 14). This directional solvent flow entrains a high concentration of dissolved FAI towards the liquid–air interface, thereby inducing supersaturation and subsequent precipitation. Crucially, this directional co-transport is greatly enhanced by the strong hydrogen-bonding interactions between 2-Me and FAI (Supplementary Fig. 15), providing a mechanistic rationale for the preferential surface enrichment.

The ternary high-SVP solvent system (2-Me:DOL:DMSO) not only ensured the efficient solvent removal and tailored perovskite surface

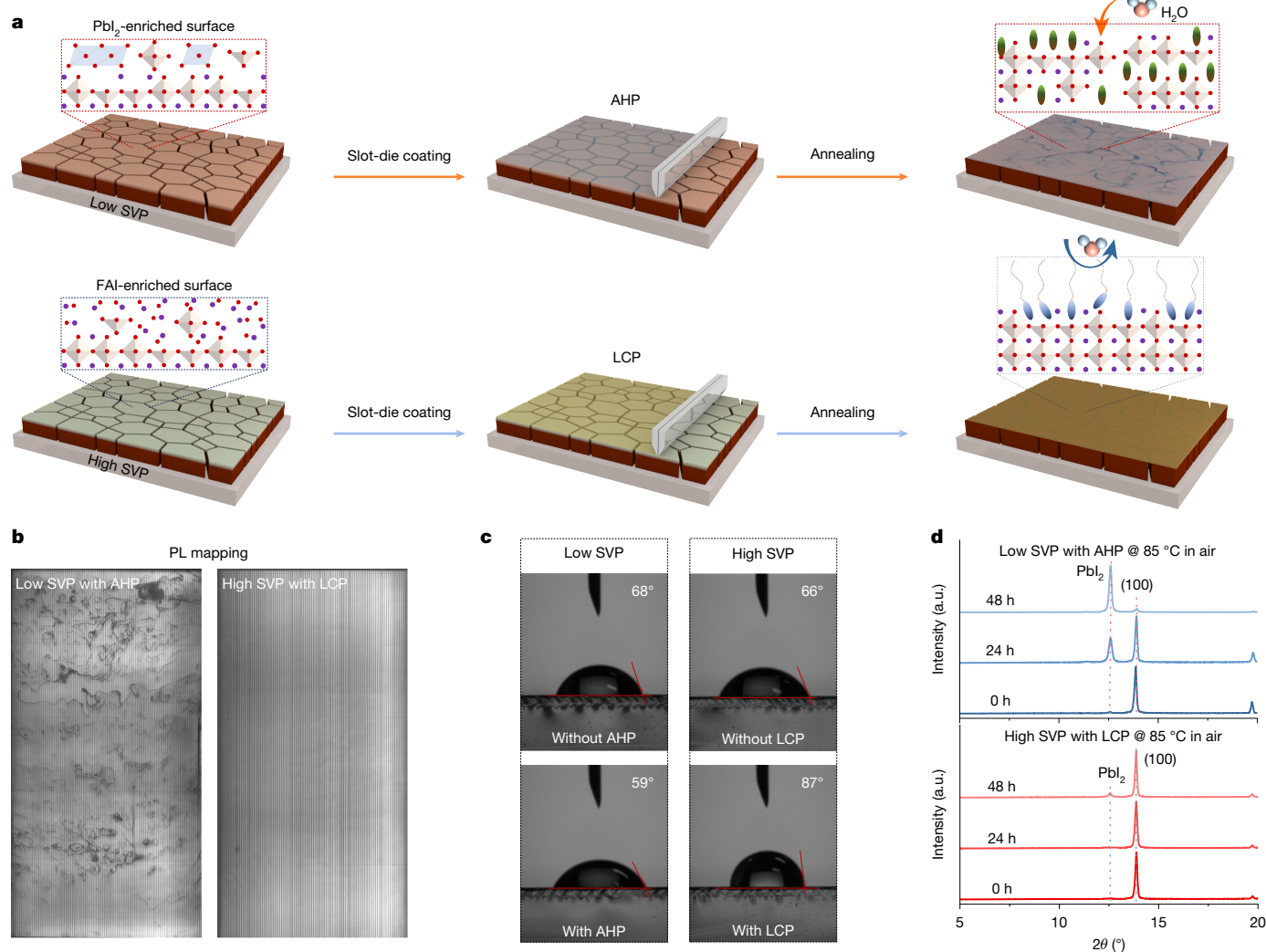


Fig. 2 | Different perovskite surface passivators. a, Schematic illustration of slot-die coating process for AHP and LCP. **b**, PL mapping of AHP and LCP commercial-scale PV modules ($1.2 \times 0.6 \text{ m}^2$). **c**, Water/perovskite film contact angle images. **d**, XRD patterns of AHP and LCP perovskite films after thermal

aging at 85 °C under ambient conditions (30–50% RH). Low SVP and high SVP refer to the solvent systems DMF:DMSO = 9:1 (volume ratio) and 2-Me:DOL:DMSO = 6:3:1, respectively; AHP and LCP denote EDAI_2 and $\text{Pb}(\text{OA})_2$, respectively. a.u., arbitrary units.

but also enabled the fabrication of large high-quality perovskite films, a critical prerequisite for efficient and stable PSMs. To determine the optimal content of DOL in the solution, we fabricated $30 \times 40\text{-cm}^2$ PSMs in a FTO/ NiO_x /perovskite/passivator/ C_{60} /ALD- SnO_2 /IZO/Cu architecture (Supplementary Note 2 and Supplementary Figs. 16–21). Addition of DOL into the perovskite solvent system led to improvements in PSM performance: the low-SVP (DMF:DMSO), 2-Me:DMSO and high-SVP (2-Me:DOL:DMSO) systems achieved peak PCEs of 19.0%, 20.6% and 22.5%, respectively, mainly driven by increases in open-circuit voltage (V_{oc}) and fill factor (FF) (Supplementary Figs. 22 and 23 and Supplementary Table 2).

As observed from the scanning electron microscopy (SEM) images, the perovskite film prepared using the tailored high-SVP solvent system exhibited well-defined grains and improved morphology on both surfaces (Supplementary Fig. 24). X-ray diffraction (XRD) patterns showed that perovskite film prepared using high SVP exhibited stronger (100) crystallographic orientation than those fabricated from low-SVP solvents, which also showed a distinct PbI_2 phase (Supplementary Fig. 25). Grazing incidence wide-angle X-ray scattering (GIWAXS) at 0.4° indicated that the perovskite film fabricated using the high-SVP solvent showed dominant diffraction features associated with the (100) plane, accompanied by reduced diffuse scattering³⁸. By contrast,

the film formed with the low-SVP solvent showed notable PbI_2 residue and the presence of the photo-inactive δ -phase perovskite (Fig. 1e and Supplementary Fig. 26). The high-SVP perovskite film exhibits stronger photoluminescence (PL) intensity and longer photogenerated carrier lifetime (Supplementary Figs. 27 and 28). Furthermore, introducing excess FAI into the low-SVP system fails to replicate the performance exhibited by the high-SVP approach (Supplementary Note 3 and Supplementary Figs. 29–33), further suggesting the necessity and superiority of the proposed LCP method.

Selection of perovskite surface passivators

Passivators for solution-processed large-area perovskite PV modules can be deposited by slot-die coating (Fig. 2a and Supplementary Fig. 34). AHPs, such as phenethylammonium iodide (PEAI)^{39,40}, phenethylammonium bromide (PEABr)^{41,42}, propane-1,3-diammonium diiodide (PDAl)^{43,44} and ethylenediammonium diiodide (EDAI_2)⁴⁵, have been routinely used for this purpose, substantially improving the performance of perovskite PVs. The function of these AHPs lies in their reactivity with PbI_2 -rich perovskite surfaces formed in conventional low-SVP (DMF and DMSO) solvent-based processing¹ (Supplementary Fig. 35a). By contrast, the use of the high-SVP system (2-Me, DOL

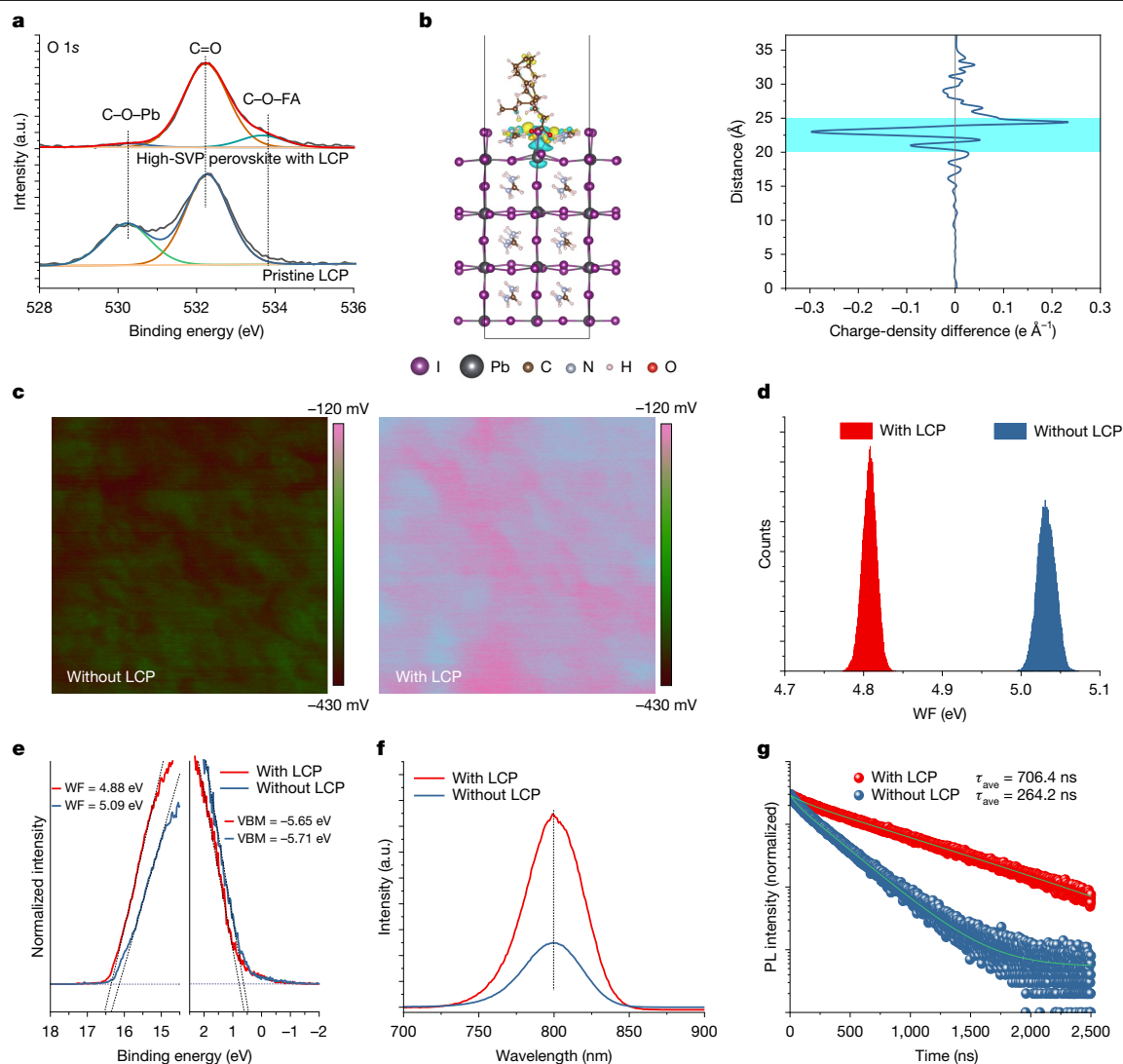


Fig. 3 | Surface passivation by LCP. **a**, O 1s XPS spectra of pristine LCP and high-SVP perovskite film with LCP. **b**, Computational models of OA[−] on a defective PbI₂-terminated surface (I[−] vacancy). Left, molecular assembly with electron depletion (blue) and accumulation (yellow). Right, interfacial charge-density difference, with the cyan band denoting the interface. **c**, KPFM surface

potential maps on perovskite films. The area is $5 \times 5 \mu\text{m}^2$. **d**, Statistical distribution of WFs from panel **c**. **e**, UPS spectra of perovskite films with and without LCP. **f, g**, Steady-state PL spectra (**f**) and TRPL decay curves (**g**) of the perovskite films with and without LCP. a.u., arbitrary units.

and DMSO), as discussed above, leads to FAI-rich surface, thereby creating a prerequisite for subsequent passivation with LCPs (Supplementary Fig. 35b). We evaluated several top-surface passivators for PSMs and identified Pb(OA)₂ as optimal for FAI-enriched perovskite surfaces formed from the high-SVP system (hereafter referred to as 'LCP') and EDAl₂ as optimal for PbI₂-enriched perovskite surfaces from the low-SVP system (hereafter referred to as 'AHP') (Supplementary Figs. 36–39 and Supplementary Tables 3 and 4).

We then examined the perovskite films treated with LCP and AHP. Unlike LCP, the AHP exhibited ring-like deposits at the edge of the wet film owing to the coffee-ring effect or island-like non-uniform morphologies during solvent evaporation, manifesting as localized macroscale defects (Supplementary Note 4 and Supplementary Fig. 40). PL mapping of a 0.72-m² PSM revealed notable heterogeneity in AHP (Fig. 2b), unlike the uniform radiative emission profile in LCP-treated PSM. Perovskite films treated with the LCP showed substantial hydrophobicity, as evidenced by increased contact angle with water, indicating their ability to resist moisture penetration (Fig. 2c and Supplementary Fig. 41). Indeed, on exposing the perovskite films to ambient air at room temperature (30–50% relative humidity (RH))

for 24 h, the LCP-modified film maintained excellent crystalline integrity. By contrast, the AHP-modified film underwent severe morphological degradation, developing a honeycomb-like porous structure as a result of moisture infiltration (Supplementary Fig. 42), which accelerated the perovskite decomposition (Supplementary Fig. 43). XRD similarly revealed that the LCP-modified perovskite films exhibited excellent environmental thermal and ultraviolet stability (Fig. 2d and Supplementary Figs. 44 and 45).

We also observed a notable reduction in PSM efficiency when AHP was used to passivate perovskite films made from the ternary high-SVP solvent system (Supplementary Fig. 46). Although AHP led to an increase in V_{oc} owing to some degree of passivation, it concurrently reduced the short-circuit current (I_{sc}) and FF, suggesting the formation of an electrically resistive interface that impedes charge transport.

To investigate the passivation mechanism by LCP, XPS was first performed. Pristine LCP exhibited the O 1s binding energy at 530.26 eV (C–O–Pb) and 532.25 eV (C=O). Following its deposition on FAI-enriched perovskite surface, a new peak emerged at 533.79 eV, which we attributed to newly formed C–O–FA bond (Fig. 3a). Furthermore, the binding energies of Pb 4f in pristine LCP shifted from 137.98 eV

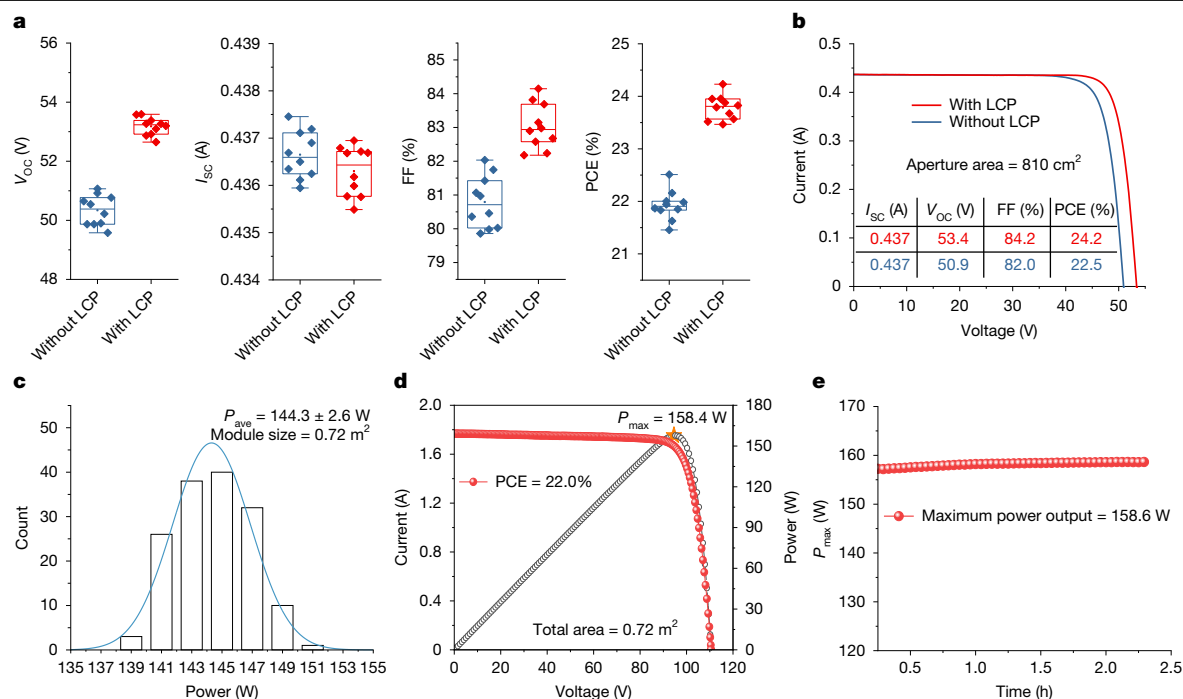


Fig. 4 | Performance of PSMs. **a**, Statistical comparison of PV performance for PSMs with and without LCP ($n = 10$ per group, aperture area = 810 cm^2). Scatters represent raw data; box plots show the interquartile range with the median as the centre line; whiskers indicate the 5th–95th percentiles. **b**, I - V curves of the best-performing PV modules. Inset table shows the PV parameters of the

corresponding curves. **c**, Statistical PV performance of LCP-modified PSMs with a total area of 0.72 m^2 . **d**, I - V curves of the best-performing LCP-modified modules with a total area of 0.72 m^2 . **e**, Stabilized P_{max} plots for the best-performing module over a period exceeding 2 h.

and 142.82 eV to 138.91 eV and 143.71 eV, respectively, on deposition on the FAI-enriched perovskite surface (Supplementary Fig. 47), indicating the formation of a more stable chemical environment for the LCP-passivated perovskite surface^{22,46}.

Density functional theory (DFT) calculations indicated that OA^- mainly adsorbs at iodine vacancy sites on the perovskite surface to coordinate with undercoordinated lead and simultaneously binds to FA^+ cations through hydrogen bonding (Fig. 3b). This adsorption configuration exhibits a higher binding energy than that of I^- at the same sites (Supplementary Fig. 48). The distribution of OA^- in the three-dimensional depth profile of COO^+ , as measured by TOF-SIMS, showed that the oleate was predominantly localized at the film surface, with only trace amounts penetrating into the perovskite bulk by means of grain boundaries (Supplementary Fig. 49). The measured activation energy (E_a) and cross-sectional potential difference of the perovskite film confirm that this passivation strategy does not adversely affect ion migration within the film (Supplementary Note 5 and Supplementary Figs. 50–52).

To further understand the interaction between the LCP and the FAI-enriched perovskite surface, we investigated the reaction between the LCP and FAI and found that a characteristic XRD peak of α -FAPbI₃ (13.9°) emerged only at a FAI:LCP ratio of 3:1 (Supplementary Fig. 53). With increasing FAI content, the mixture experienced phase evolution from PbI₂ and δ -FAPbI₃ to α -FAPbI₃. The measured bandgap and transmission electron microscopy of the 3:1 film further confirmed the formation of the α -FAPbI₃ (Supplementary Figs. 54 and 55). We hence propose that the LCP reacts with the FAI to form nanosized α -FAPbI₃ perovskite crystals capped with outward-oriented alkyl chains (Supplementary Fig. 56), with OA^- ligands simultaneously passivating both top and inter-grain surfaces (Supplementary Fig. 57).

The top-view SEM and atomic force microscopy (AFM) images demonstrated a smoother surface morphology after the treatment of FAI-surface-enriched perovskite by LCP (Supplementary Figs. 58 and 59). Kelvin probe force microscopy (KPFM) revealed a more uniform surface potential distribution on the LCP-modified perovskite film, with the

average potential increasing from -395 mV to -168 mV (Fig. 3c and Supplementary Fig. 60). The calculated work function (WF) distribution for the LCP-modified perovskite film was 4.78–4.83 eV, notably lower than that of the untreated film (4.99–5.07 eV) (Fig. 3d). These results demonstrate that LCP modification reduces surface energy-level inhomogeneity, optimizes interfacial energy-level alignment and thereby enhances charge extraction⁴⁷.

Ultraviolet photoelectron spectroscopy (UPS) was used to determine the energy-band level of perovskite films. Compared with the control perovskite film with WF of 5.09 eV and valence band maximum (VBM) of -5.71 eV , the LCP-modified film showed a lower WF of 4.88 eV and a shallower VBM of -5.65 eV (Fig. 3e). The energy-level diagram of the LCP-passivated perovskite, constructed using the conduction band minimum (CBM) determined from optical bandgap measurements, shows a shallower CBM, which enables more favourable band alignment with C_{60} and hence enhanced electron extraction¹ (Supplementary Fig. 61).

To gain insight into charge-carrier recombination kinetics, we performed PL and time-resolved photoluminescence (TRPL) spectroscopy. The LCP-modified perovskite film showed a greatly enhanced PL intensity (Fig. 3f). Correspondingly, its average carrier lifetime increased from 264 ns to 706 ns (Fig. 3g). This improvement was corroborated by the carrier lifetime mapping (Supplementary Fig. 62). Transient absorption spectroscopy further confirmed the improved carrier lifetime in the LCP-modified perovskite films (Supplementary Fig. 63). At a probe wavelength of 765 nm, both films exhibited ground state bleaching. However, the LCP-modified film showed a notably slower ground state bleaching decay, indicating prolonged carrier lifetime¹⁴. These results indicate that the LCP effectively reduced carrier trapping in the perovskite^{48,49}.

Performance of PSMs

To examine the impact of the optimized LCP on device performance, we fabricated $30 \times 40\text{-cm}^2$ PSMs with an aperture area of 810 cm^2 and

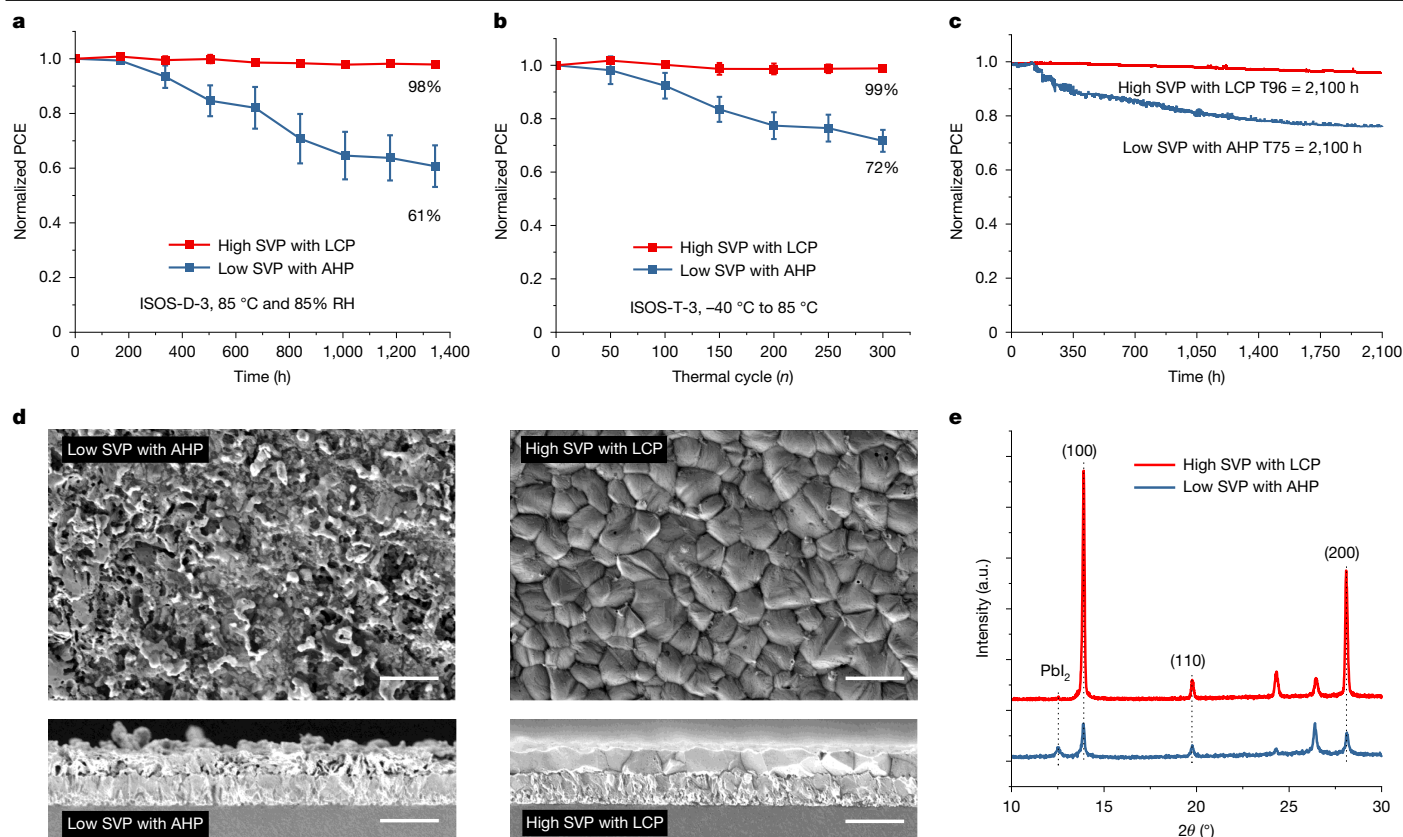


Fig. 5 | Reliability tests of 0.72-m² PV modules. **a, b**, Encapsulated PV modules aged under ISOS-D-3 (**a**) and ISOS-T-3 (**b**) testing protocols (five modules for each condition). **c**, MPPT of the encapsulated modules under simulated AM1.5G illumination in ambient air with a RH of 30–50%. The modules reached 60 to 65 °C during the test. **d**, Top-view (upper) and cross-sectional (lower) SEM

images of perovskite films from modules subjected to the ISOS-D-3 ageing test (conditions shown in panel **a**). Scale bars, 1 μm . **e**, XRD patterns of perovskite films from modules subjected to the ISOS-D-3 ageing test (conditions shown in panel **a**). a.u., arbitrary units.

compared their PV parameters. The efficiency of PSMs fabricated using low-SVP solvents can be further improved through optimal passivation with AHP, whereas those fabricated using high-SVP solvents can be optimized with LCP (Supplementary Figs. 64 and 65 and Supplementary Table 5). The incorporation of LCP greatly improved the performance of PSMs (Fig. 4a), particularly in both V_{oc} and FF, as expected from effective passivation. A champion efficiency of 24.2% was obtained with the LCP (Fig. 4b), featuring a V_{oc} of 53.4 V, an I_{sc} of 0.437 A and a FF of 84.2%. Simultaneously, a stable power output of 19.4 W (corresponding to a PCE of about 24.0%) was maintained under maximum power point tracking (MPPT) for 20 min (Supplementary Fig. 66). By comparison, the best-performing PSM without passivation exhibited lower V_{oc} and FF, resulting in an efficiency of 22.5%. An independent certification yielded an efficiency of 24.0% (I_{sc} = 0.436 A, V_{oc} = 53.46 V, FF = 83.40%; Supplementary Fig. 67). This result represents the first report of a PSM with an aperture area exceeding 800 cm² to exceed an efficiency of 24%. Moreover, this strategy demonstrates a similarly positive effect on efficiency in wide-bandgap perovskite systems (1.68 eV and 1.80 eV) (Supplementary Figs. 68–71), an extension that substantially enhances the generality and impact of the conclusions presented in this work.

We then fabricated 150 commercial-sized PSMs with 0.72-m² area (120 × 60 cm). The corresponding PCE histogram in Fig. 4c revealed an average power output of 144 W. The champion module achieved a certified PCE of 22.0% (158.4 W), with a V_{oc} of 110.8 V, I_{sc} of 1.77 A and FF of 81.0% under reverse scan (Fig. 4d and Supplementary Fig. 72). Furthermore, the module exhibited stable power output over a period exceeding 2 h under MPPT, reaching a maximum power output of 158.6 W (Fig. 4e). This result represents the highest reported PCE for metre-scale PSMs so far.

Reliability of commercial-scale PV modules

Beyond efficiency, stability is another key metric for commercial PV modules. We first demonstrated that the LCP strategy fulfils the requirement for tolerance to air exposure during fabrication (Supplementary Fig. 73) and subsequently investigated the efficiency degradation and morphological evolution of PSMs fabricated with AHP and LCP under IEC-standard testing conditions (Supplementary Note 6). After 1,300 h of damp-heat testing (85% RH/85 °C, ISOS-D-3), AHP-based modules lost 39% of their initial efficiency (relative), whereas the LCP-modified modules exhibited only 2% relative loss (Fig. 5a and Supplementary Fig. 74). After 300 thermal cycles (between –40 °C and 85 °C, ISOS-T-3), the LCP-modified modules exhibited negligible efficiency loss (Fig. 5b and Supplementary Fig. 75). The LCP-modified modules retained 96% of their initial efficiency after 2,200 h under MPPT (Fig. 5c) and 95% of their initial performance under ultraviolet ageing for a total solar irradiance of 30 kWh m^{–2} (Supplementary Fig. 76). By contrast, modules with AHP modifications consistently exhibited inferior ageing tolerance. Moreover, the efficiency and stability trends of the modules closely mirrored those of their constituent unit cells (Supplementary Note 7, Supplementary Figs. 77–83 and Supplementary Tables 6–11). We further evaluated the PCEs and reliability of AHP systems exhibiting hydrophobicity comparable with that of LCP (Supplementary Note 8 and Supplementary Figs. 84–89), thereby validating the advantages of the LCP strategy.

For morphological characterization, the aged PV modules subjected to ISOS-D-3 testing were delaminated by mechanically scribing and fracturing the structure, thereby removing the overlying C₆₀/ALD-SnO₂/IZO/Cu electrode stack to reveal the underlying perovskite layer. SEM imaging clearly revealed that the perovskite grains under AHP

treatment underwent marked decomposition, whereas those modified with LCP maintained intact grain geometries (Fig. 5d). Consistently, XRD further confirmed retention of crystallographic orientation in the LCP-modified samples (Fig. 5e).

All reliability tests of the PSMs with LCP modification exceeded the requirements of the IEC 61215 standard, demonstrating that the tailored passivation strategy endows the perovskite surface with enhanced physicochemical stability, improving the environmental tolerance of the modules. We conducted a three-month monitoring campaign (March to May 2026) on a 1-MW PV system and a 3.5-MW silicon (TOP-Con) PV system, both installed within the same utility-scale power plant (Supplementary Fig. 90). A comparative analysis of the specific yield over the monitoring period confirmed that all systems delivered stable power output. However, the perovskite module consistently outperformed its silicon counterpart in terms of daily specific energy yield. The monthly specific yield gain ratios for March, April and May were 3.42%, 3.79% and 5.81%, respectively, demonstrating superior real-world performance (Supplementary Fig. 91). These results validate the commercial viability of PV technology.

By integrating a high-SVP solvent system with a chemically stable LCP, we demonstrate a robust, industry-ready protocol for fabricating large-area perovskite modules at the metre scale. Crucially, this strategy not only achieves state-of-the-art performance but also passes the full suite of IEC 61215 qualification tests. Our findings establish a new benchmark for manufacturable PVs, in which processability, durability and efficiency are no longer mutually exclusive but are engineered from the outset. This work thus charts a clear course towards the industrialization of perovskite solar technology.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-026-10994-7>.

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Methods

Materials

DMF (anhydrous, 99.8%), 2-Me (anhydrous, 99.8%), DMSO (anhydrous, 99.9%), 2-propanol (IPA, anhydrous, 99.5%) and DOL (anhydrous, 99.8%) were purchased from Sigma-Aldrich. CsI (99.999%), PbI_2 (99.99%) and Me-4PACz were purchased from TCI. FAI (99.9%) and methylammonium chloride (MACl, 99.9%) were purchased from GreatCell Solar. C_{60} (fullerene) was purchased from Nano-C. The AHPs (PEAI; PEABr; ethanediaminedihydroiodide, EDAl₂; PDAI₂) were purchased from Xi'an Yuri Solar Co., Ltd. The organolead salts passivators (lead acetate, $\text{Pb}(\text{Ac})_2$; lead naphthenate, $\text{Pb}(\text{NA})_2$; lead acetylacetonate, $\text{Pb}(\text{AcAc})_2$; $\text{Pb}(\text{OA})_2$; trimethyl-tetradecylammonium chloride, TAC; oleylamine hydrochloride, OAC) were purchased from Aladdin and Energy Chemical. All chemicals were used as received without further purification.

$\text{Cs}_{0.03}\text{FA}_{0.97}\text{PbI}_3$ perovskite ink (1.54 eV)

The perovskite powders including 233.8 mg CsI, 5,005.2 mg FAI, 13,830.0 mg PbI_2 and 324.1 mg MACl were dissolved in mixed solvent (40 ml). The low-SVP solvent is composed of a mixture of DMF and DMSO (9:1, volume/volume). The high-SVP solvent is composed of a mixture of 2-Me, DOL and DMSO (6:3:1, volume/volume/volume). The ink was stirred at room temperature for at least 1 h and then filtered through a 0.22- μm polytetrafluoroethylene (PTFE) membrane before deposition.

$\text{Cs}_{0.05}\text{MA}_{0.05}\text{FA}_{0.90}\text{Pb}(\text{I}_{0.75}\text{Br}_{0.25})_3$ perovskite ink (1.68 eV)

The perovskite powders including 312 mg CsI, 3,715 mg FAI, 191 mg MAI, 6,915 mg PbI_2 , 3,307 mg PbBr_2 and 324 mg MACl were dissolved in mixed solvent (40 ml). The low-SVP solvent is composed of a mixture of DMF and DMSO (8:2, volume/volume). The high-SVP solvent is composed of a mixture of 2-Me, DOL and DMSO (5:3:2, volume/volume/volume). The ink was stirred at room temperature for at least 1 h and then filtered through a 0.22- μm PTFE membrane before deposition.

$\text{Cs}_{0.15}\text{FA}_{0.85}\text{Pb}(\text{I}_{0.6}\text{Br}_{0.4})_3$ perovskite ink (1.80 eV)

The perovskite powders including 936 mg CsI, 3,509 mg FAI, 4,426 mg PbI_2 , 5,285 mg PbBr_2 and 162 mg MACl were dissolved in mixed solvent (40 ml). The low-SVP solvent is composed of a mixture of DMF and DMSO (8:2, volume/volume). The high-SVP solvent is composed of a mixture of DMF, 2-Me, DOL and DMSO (1:4:3:2, volume/volume/volume/volume). The ink was stirred at room temperature for at least 1 h and then filtered through a 0.22- μm PTFE membrane before deposition.

Testing, characterization and measurements

To evaluate the surface tension properties of the perovskite precursor solution, static contact angle measurements were conducted using a contact angle goniometer (DataPhysics OCA 15EC) at room temperature. For the contact angle measurements, droplets of the water (approximately 100 μl) were deposited onto the perovskite film surface and the contact angle was measured using the sessile drop method by an optical microscope (Olympus BX43, 10 $\times$ objective, bright-field mode). The current–voltage (I – V) characteristics of the devices were characterized under simulated solar illumination using a calibrated LED-based composite light source (GSIV-22A1006LED) at an irradiance of 1,000 W m^{-2} in ambient air. The film quality was examined using a field-emission scanning electron microscope (Rigaku, 5 kV), with high-resolution images acquired through an in-lens detector. Steady-state PL and TRPL spectra were acquired using a Horiba Fluorolog spectrofluorometer equipped with time-correlated single-photon-counting capabilities. XRD measurements were performed using a Bruker D8 Advance diffractometer in the 2θ range 5–45°.

The surface morphology of the samples was characterized using AFM (Bruker Dimension Icon) in tapping mode. A silicon cantilever with a nominal spring constant of 40 N m^{-1} and a resonant frequency of 300 kHz was used to obtain high-resolution topographic images. The root-mean-square roughness was calculated from the AFM data over a scan area of $5 \times 5 \mu\text{m}^2$. The KPFM measurements were performed on a Bruker Dimension Icon atomic force microscope in an argon-filled glovebox. We used a Pt–Ir-coated PPP-EFM probe in tapping mode. Nuclear magnetic resonance measurement was conducted by a Bruker DPX 400 MHz. The magnetic field intensity was 9.4 T (400 MHz for ^1H). The system was equipped with a fast-entry load-lock, a transfer chamber and an analysis chamber (base pressure approximately 10–10 mbar). UPS used He I (21.22 eV) as the excitation source with an energy resolution of 50 meV. The spectra were calibrated by referring the Fermi level (E_F) to the Au $4f_{7/2}$ position of the Ar⁺ ion sputter-cleaned Au foil.

Calculations

AIMD simulations. Ab initio molecular dynamics (AIMD) simulations were performed using the CP2K package. The simulation model consisted of a mixture of two Me molecules, one DOL molecule and one Cs⁺ ion. All AIMD simulations were conducted in the constant-volume and constant-temperature (NVT) ensemble for a total duration of 8 ps. The system temperature was maintained at 300 K using a Nosé–Hoover thermostat, with the pressure set to 1 bar. A time step of 1.0 fs was used throughout the simulations. Electronic exchange–correlation effects were described using the Perdew–Burke–Ernzerhof functional augmented with D3 dispersion corrections. Double-zeta valence polarized (DZVP-MOLOPT) basis sets were used in combination with Goedecker–Teter–Hutter pseudopotentials and the plane-wave cut-off energy was set to 400 Ry.

DFT. First-principles calculations based on DFT were performed using the Vienna Ab initio Simulation Package (VASP). The exchange–correlation interactions were described using the Perdew–Burke–Ernzerhof functional and long-range van der Waals interactions were incorporated using the DFT-D3 correction scheme. A plane-wave basis set with a kinetic energy cut-off of 400 eV was used. Structural relaxations were performed until the total energy and forces converged to within 10^{-5} eV and $0.03 \text{ eV } \text{\AA}^{-1}$, respectively.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data that support the findings of this study are available in the paper and its Supplementary Information. Source data are provided with this paper.

Author contributions H.T. and K.X. conceived the idea and supervised the overall project. D.X., Y.L., X. Luo, K.X. and Q.H. fabricated all modules. J.X., Y. Wei and S.Y. performed the calculations. D.X. designed and performed the module characterization. H.L. participated in the XRD measurements and drew the schematic illustrations. Y. Wang, S.H. and M.L. assisted with the GIWAXS, PL and TRPL characterizations. Y.Z., H.G., H.S. and G.D. contributed to the TOF-SIMS, XPS and UPS measurements and data analysis. Q.L., S.Z., X. Lv, J.J., L.C., D.L., M.H. and X.H. constructed the PV power station. Y.A. and M.I.S. provided analysis on the cosolvency of DOL:2-Me solvents and polished the manuscript. D.X. wrote and revised the paper. All authors discussed the results and commented on the paper.

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Competing interests Patent applications (CN202511007795.3 and CN202511007664.5) have been filed by Renshine Solar (Suzhou) Co., Ltd. based on this work. H.T. is the founder, chief scientific officer and chairman of Renshine Solar (Suzhou) Co., Ltd., a company that is commercializing perovskite PVs. The other authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41586-026-10994-7>.

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Solar Cells Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form is intended for publication with all accepted papers reporting the characterization of photovoltaic devices and provides structure for consistency and transparency in reporting. Some list items might not apply to an individual manuscript, but all fields must be completed for clarity.

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► Experimental design

Please check the following details are reported in the manuscript, and provide a brief description or explanation where applicable.

1. Dimensions

Area of the tested solar cells

☒ Yes
☐ No

We characterized two types of modules: small-area modules with an aperture area of 810 cm² and commercial-size modules with a total area of 0.72 m².

Explain why this information is not reported/not relevant.

Method used to determine the device area

☒ Yes
☐ No

Black metal aperture masks were used during the J-V measurements of modules with an aperture area of 810 cm².

Explain why this information is not reported/not relevant.

2. Current-voltage characterization

Current density-voltage (J-V) plots in both forward and backward direction

☒ Yes
☐ No

Provided in this manuscript.

Voltage scan conditions

☒ Yes
☐ No

The sweeping conditions are:

For the modules with an aperture area of 810 cm²: Scan range (-5 V → 75 V, a scan rate of 4 V s⁻¹, and delay time 10 ms);

For the commercial-size modules with a total area of 0.72 m²: Scan range (-5 V → 150 V, a scan rate of 10 V s⁻¹, and delay time 10 ms);

Explain why this information is not reported/not relevant.

Test environment

☒ Yes
☐ No

The modules were all measured in ambient air.

Explain why this information is not reported/not relevant.

Protocol for preconditioning of the device before its characterization

☐ Yes
☒ No

Provide a description of the protocol.

No preconditioning was used in this work.

Stability of the J-V characteristic

☒ Yes
☐ No

Stabilized PCEs of the modules were provided in Figure 4e and Supplementary Figure 66.

Explain why this information is not reported/not relevant.

3. Hysteresis or any other unusual behaviour

Description of the unusual behaviour observed during the characterization

☒ Yes
☐ No

Minor hysteresis was observed for devices herein.

Explain why this information is not reported/not relevant.

Related experimental data

☒ Yes
☐ No

Related J-V curves and data were provided in Supplementary Figure 67 and Supplementary Figure 72.

Explain why this information is not reported/not relevant.

4. Efficiency

External quantum efficiency (EQE) or incident photons to current efficiency (IPCE)

☐ Yes
☒ No

Provide a description of the technique used.

Not provided in the manuscript.

A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator	☐ Yes ☒ No	State where this information can be found in the text. Not provided in the manuscript.
For tandem solar cells, the bias illumination and bias voltage used for each subcell	☐ Yes ☒ No	Provide a description of the measurement conditions. Tandem solar cells are not considered in this article.
5. Calibration		
Light source and reference cell or sensor used for the characterization	☒ Yes ☐ No	The current-voltage (I-V) characteristics of the devices were characterized under simulated solar illumination using a calibrated LED-based composite light source (GSIV-22A1006LED) at an irradiance of 1000 W m ⁻² in ambient air. Explain why this information is not reported/not relevant.
Confirmation that the reference cell was calibrated and certified	☒ Yes ☐ No	The reference cells were calibrated by NREL. Explain why this information is not reported/not relevant.
Calculation of spectral mismatch between the reference cell and the devices under test	☐ Yes ☒ No	Provide a value of the spectral mismatch and/or a description of how it has been taken into account in the measurements. The light spectrum used for measurements matches well with the reference silicon cell, and we did not calculate the spectral mismatch between the reference cell and the tested devices.
6. Mask/aperture		
Size of the mask/aperture used during testing	☒ Yes ☐ No	Metal aperture masks with area of 810 cm ² was used for testing. Explain why this information is not reported/not relevant.
Variation of the measured short-circuit current density with the mask/aperture area	☐ Yes ☒ No	Report the difference in the short-circuit current density values measured with the mask and aperture area. We measured all modules with an aperture area of 810 cm ² with masks.
7. Performance certification		
Identity of the independent certification laboratory that confirmed the photovoltaic performance	☒ Yes ☐ No	The modules were certified by TÜV SÜD. Explain why this information is not reported/not relevant.
A copy of any certificate(s)	☒ Yes ☐ No	The certificates were provided in Supplementary Figure 67 and Supplementary Figure 72. Explain why this information is not reported/not relevant.
8. Statistics		
Number of solar cells tested	☒ Yes ☐ No	Device number was provided in each statistics, including Figure 4a and 4c in main text, Supplementary Figure 21, 22, 31, 36, 38, 64, 68, 70, 73, 88. Explain why this information is not reported/not relevant.
Statistical analysis of the device performance	☒ Yes ☐ No	Stated in the manuscript. Explain why this information is not reported/not relevant.
9. Long-term stability analysis		
Type of analysis, bias conditions and environmental conditions	☒ Yes ☐ No	Stated in Manuscript. (MPP tracking, DH, TC, UV-test) Explain why this information is not reported/not relevant.